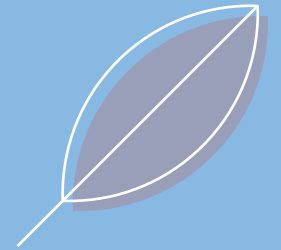


Analysis of the Iris Dataset with AWS Glue and Amazon QuickSight

Lizzy Dale

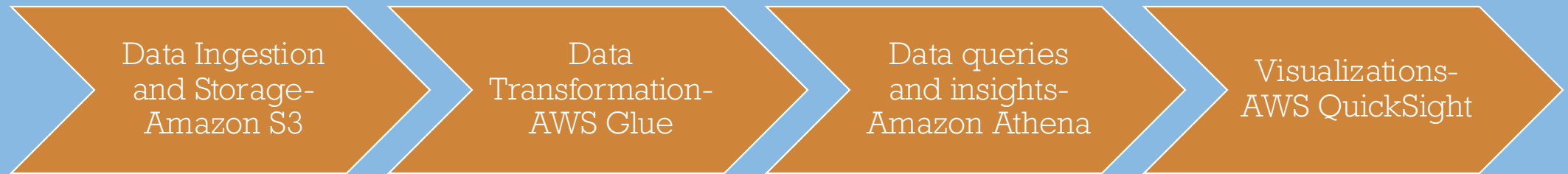


BACKGROUND

- **Overview of the Iris Dataset:**
- A prominent dataset in data science and machine learning, featuring 150 samples from three Iris species: Setosa, Versicolor, and Virginica.
- **Data Characteristics:**
- Each sample includes four measurements: sepal length, sepal width, petal length, and petal width.
- **Project Objectives:**
- Develop a data pipeline to process, analyze, and visualize the Iris dataset to identify feature correlations and classify species.



PIPELINE



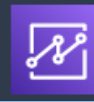
S3



AWS Glue



Cloud9



QuickSight

DATA INGESTION/STORAGE-AMAZON S3

- Data ingested into S3

iris-data-bucket333 Info

[Objects](#)[Properties](#)[Permissions](#)[Metrics](#)[Management](#)[Access Points](#)

Objects (1) Info

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

1

☐	Name	Type	Last modified	Size	Storage class
☐	iris.data	data	August 11, 2024, 06:35:49 (UTC- 07:00)	4.4 KB	Standard

DATA TRANSFORMATION-AWS GLUE

[AWS Glue](#) > [Crawlers](#) > [iris-crawler-cry](#)

iris-crawler-cry

Last updated (UTC)
August 11, 2024 at 13:54:54

Run crawler

Edit

Delete

Crawler properties

Name	IAM role	Database	State
iris-crawler-cry	AWSGlueServiceRole-iris	iris-database	READY
Description	Security configuration	Lake Formation configuration	Table prefix
-	-	-	-
Maximum table threshold			
-			

► Advanced settings

Crawler runs (1)

The list of crawler runs for this crawler.

Filter data

Filter by a date and time range

< 1 > ⚙️

	Start time (UTC) ▲	End time (UTC) ▼	Current/last duration ▼	Status ▼	DPU hours
☐	August 11, 2024 at 13:37:50	August 11, 2024 at 13:38:52	01 min 01 s	✔️ Completed	0

Edit schema JSON: iris_data_bucket333

Name	Version	Description
iris_data_bucket333	1	-

```
1 {  
2 {  
3   "Name": "sepal_length",  
4   "Type": "double"  
5 },  
6 {  
7   "Name": "sepal_width",  
8   "Type": "double"  
9 },  
10 {  
11   "Name": "petal_length",  
12   "Type": "double"  
13 },  
14 {  
15   "Name": "petal_width",  
16   "Type": "double"  
17 },  
18 {  
19   "Name": "species",  
20   "Type": "string"  
21 }  
22 }
```

AMAZON ATHENA QUERIES

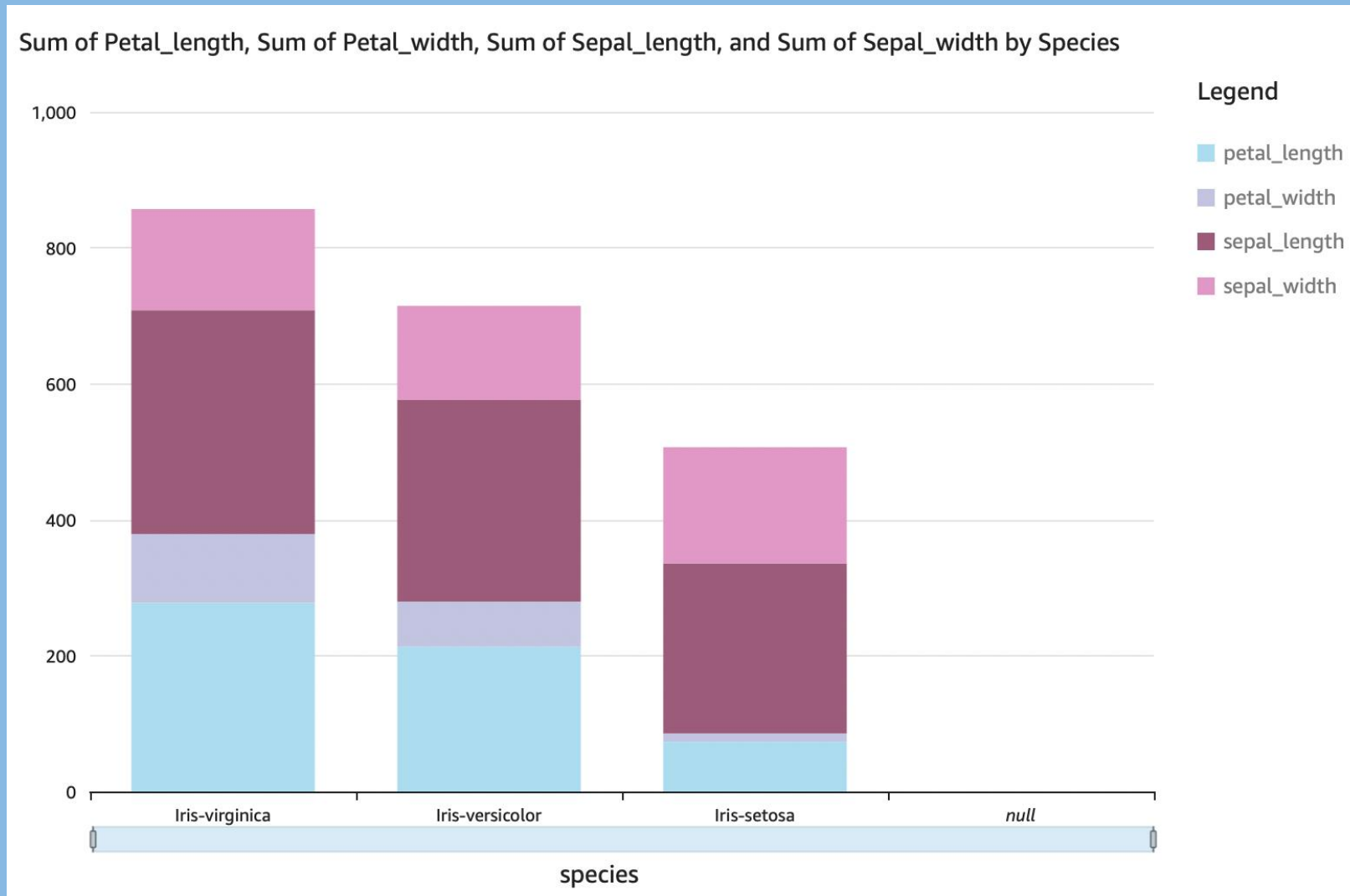
✔ Completed						Time in queue: 92 ms	Run time: 461 ms	Data scanned: 4.44 KB
Results (10)						 Copy		Download results
Search rows						< 1 > 		
# ▼	sepal_length ▼	sepal_width ▼	petal_length ▼	petal_width ▼	species ▼			
1	5.1	3.5	1.4	0.2	Iris-setosa			
2	4.9	3.0	1.4	0.2	Iris-setosa			
3	4.7	3.2	1.3	0.2	Iris-setosa			
4	4.6	3.1	1.5	0.2	Iris-setosa			
5	5.0	3.6	1.4	0.2	Iris-setosa			
6	5.4	3.9	1.7	0.4	Iris-setosa			
7	4.6	3.4	1.4	0.3	Iris-setosa			
8	5.0	3.4	1.5	0.2	Iris-setosa			
9	4.4	2.9	1.4	0.2	Iris-setosa			
10	4.9	3.1	1.5	0.1	Iris-setosa			

DESCRIPTIVE STATISTICS

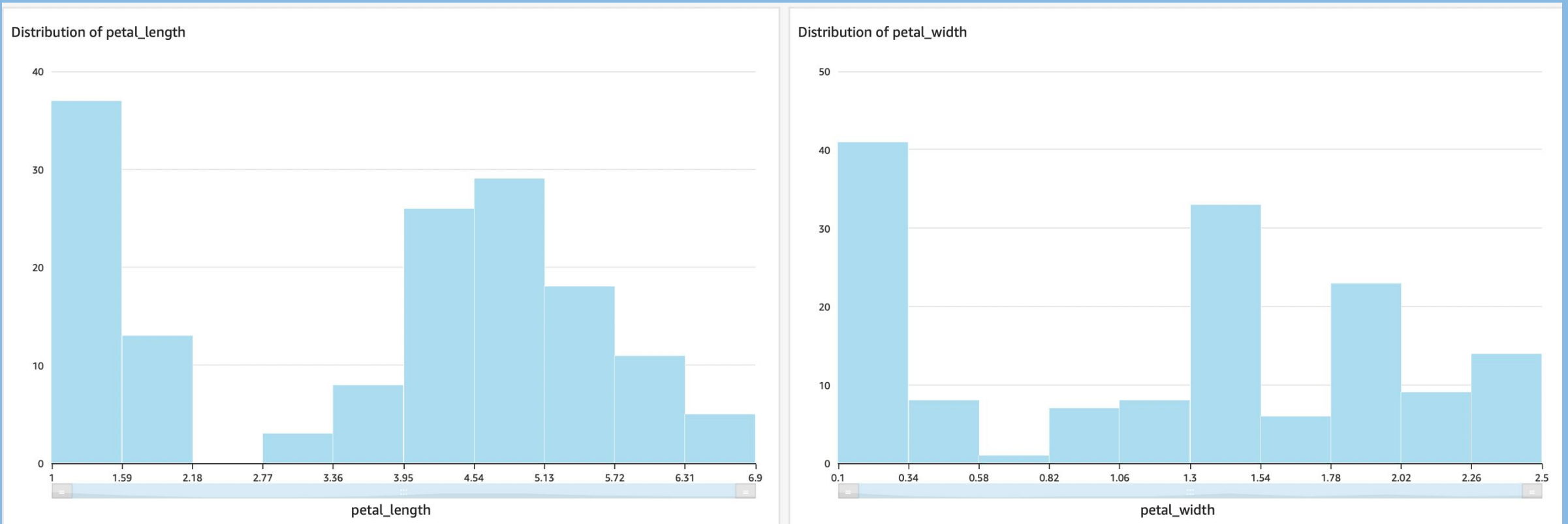
```
1 SELECT
2     species,
3     AVG(sepal_length) AS avg_sepal_length,
4     AVG(sepal_width) AS avg_sepal_width,
5     AVG(petal_length) AS avg_petal_length,
6     AVG(petal_width) AS avg_petal_width
7 FROM "iris-database".iris_data_bucket333
8 GROUP BY species;
```

#	species	avg_sepal_length	avg_sepal_width	avg_petal_length	avg_petal_width
1					
2	Iris-versicolor	5.936	2.7700000000000005	4.26	1.3259999999999998
3	Iris-virginica	6.5879999999999998	2.9739999999999998	5.552	2.026
4	Iris-setosa	5.0059999999999999	3.4180000000000006	1.464	0.24399999999999999

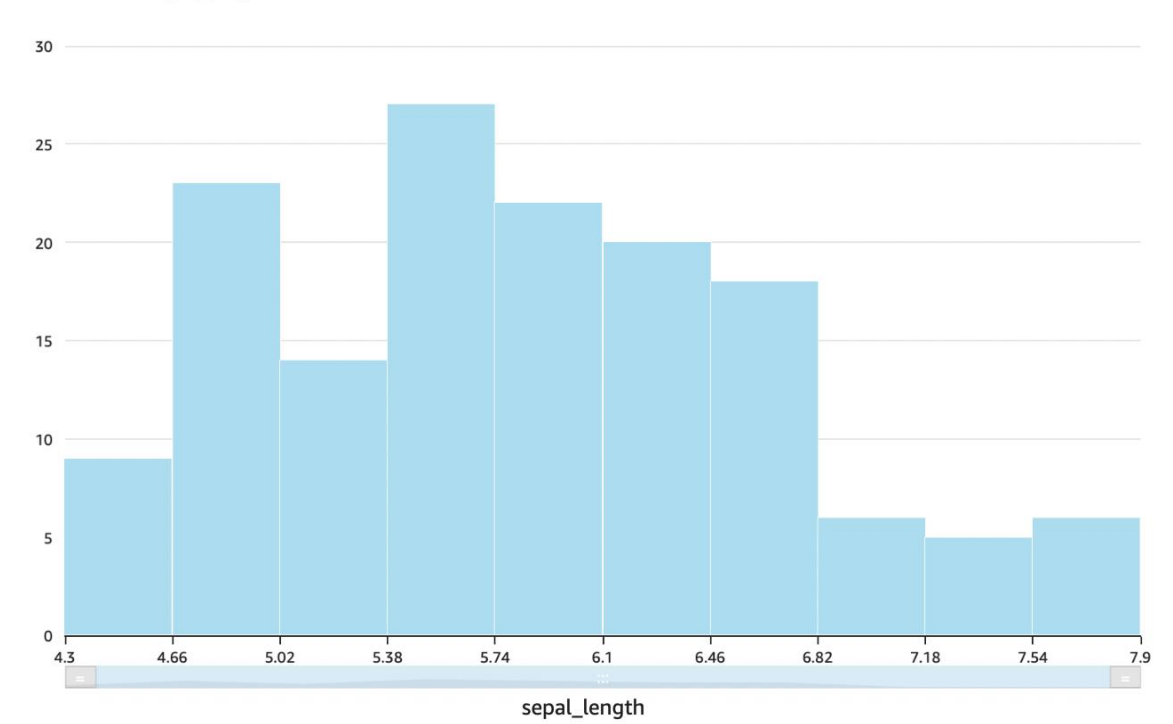
VISUALIZATIONS-AWS QUICKSIGHT



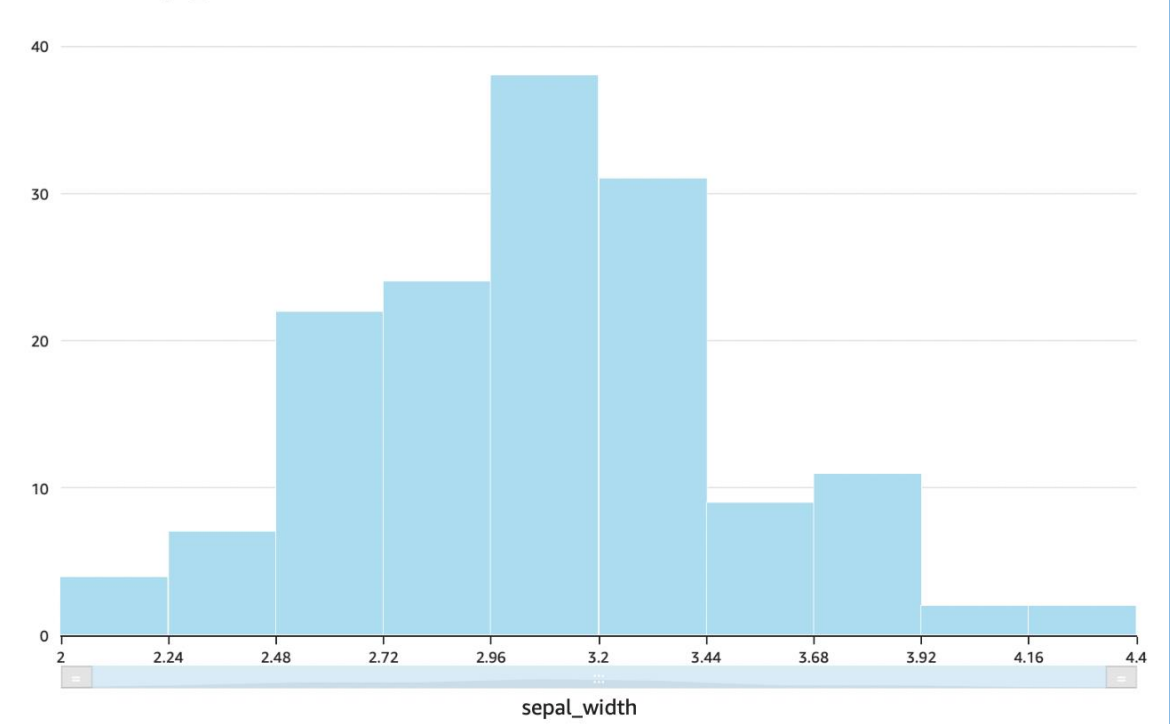
BAR PLOTS-DISTRIBUTIONS



Distribution of sepal_length



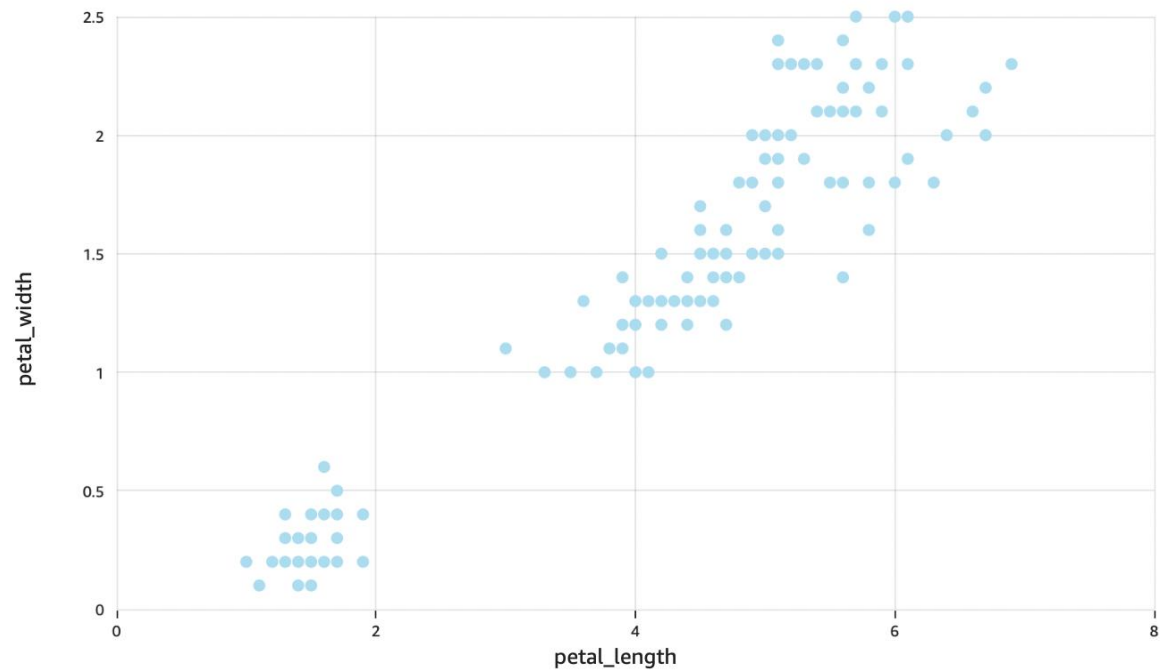
Distribution of sepal_width



SCATTER PLOTS-CORRELATIONS

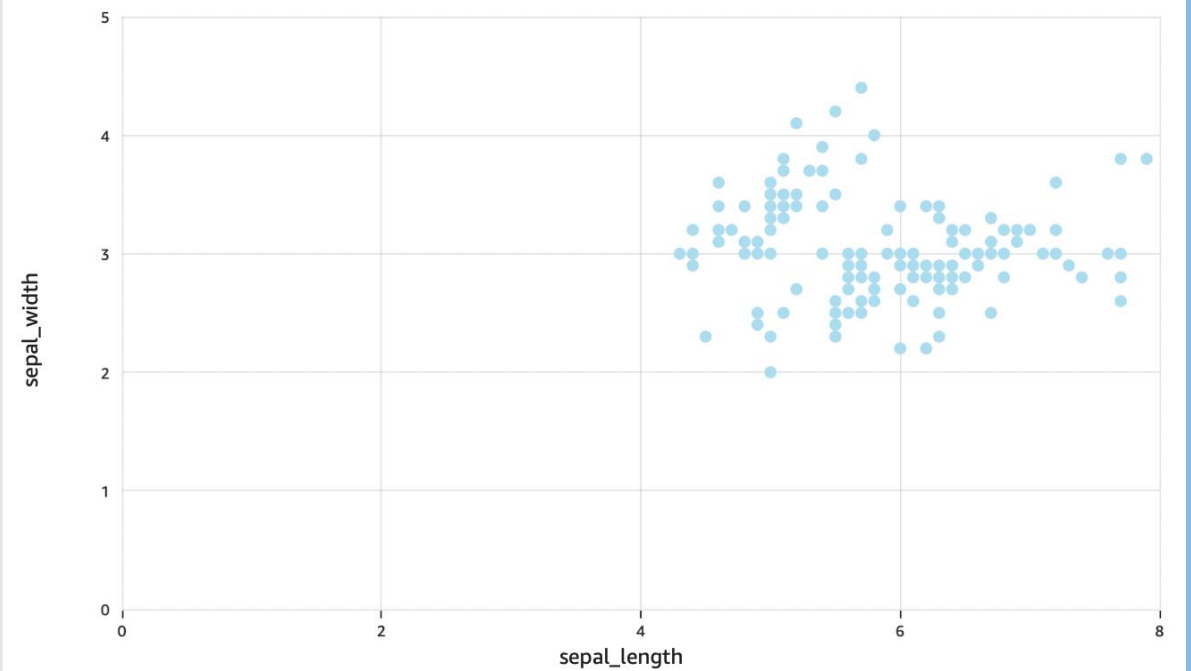
Count of Records by Petal_length and Petal_width

SHOWING UP TO 2500 DATA POINTS

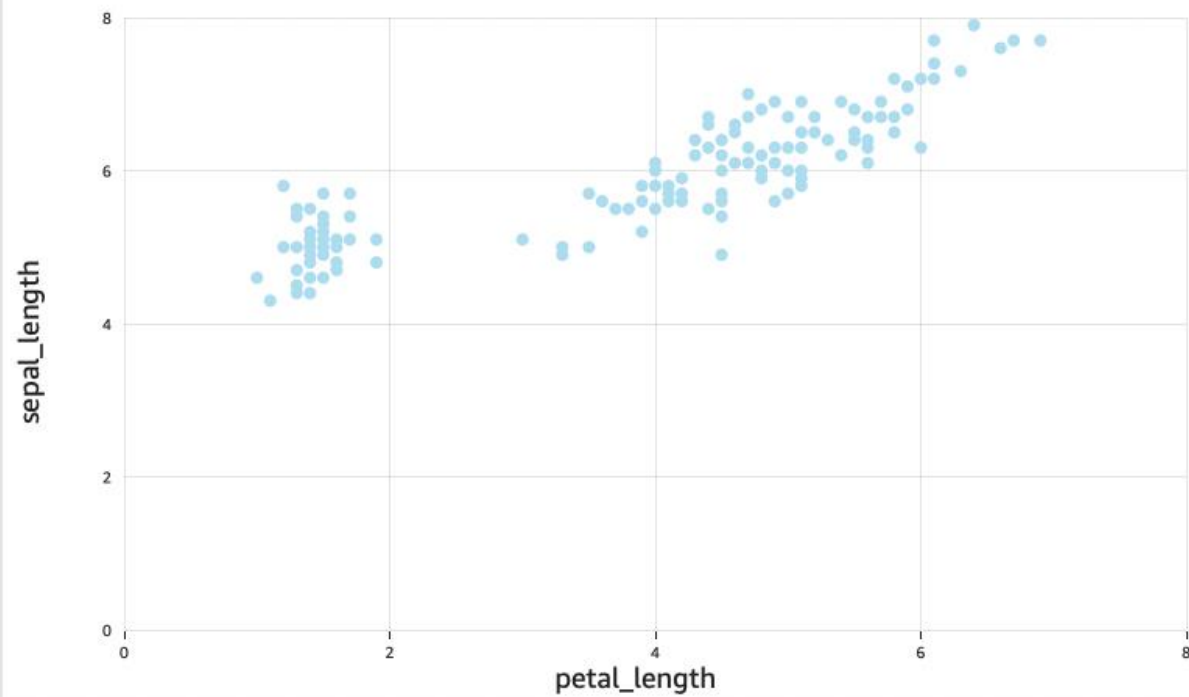


Count of Records by Sepal_length and Sepal_width

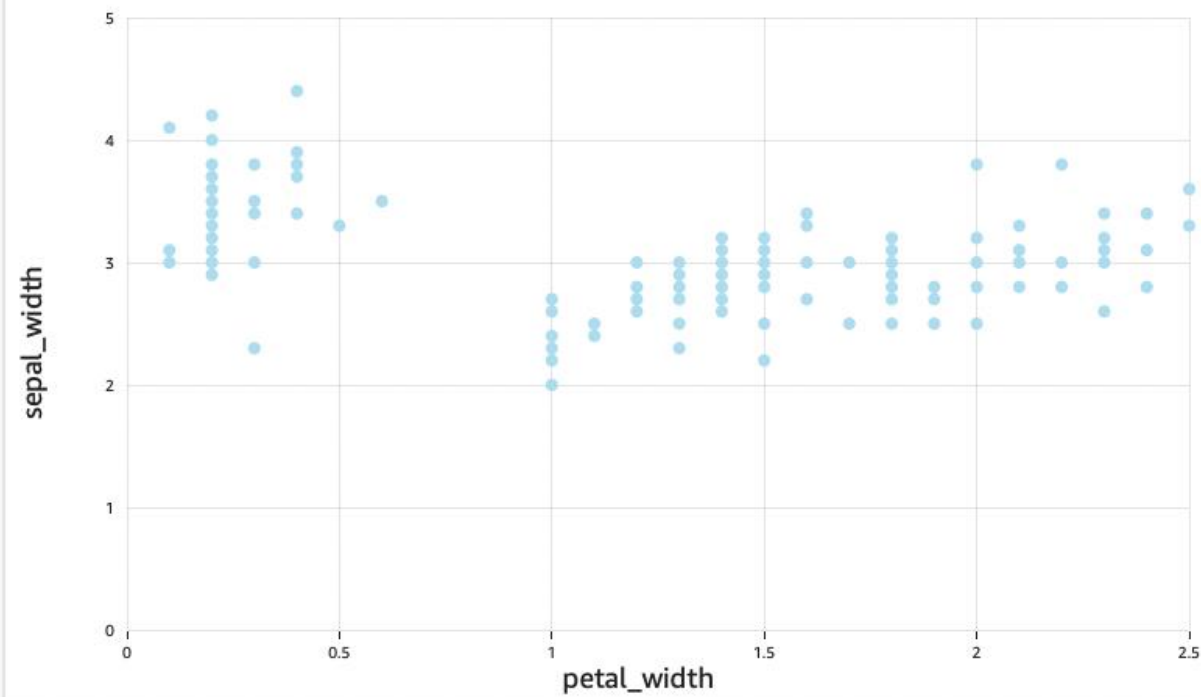
SHOWING UP TO 2500 DATA POINTS



Count of Records by Petal_length and Sepal_length
SHOWING UP TO 2500 DATA POINTS



Count of Records by Petal_width and Sepal_width
SHOWING UP TO 2500 DATA POINTS



CONCLUSIONS

- **Effective Data Handling:**
 - Utilized AWS services like S3, Glue, and Athena to efficiently collect, store, preprocess, and query the Iris dataset, demonstrating robust data management and transformation capabilities.
- **Insightful Analysis and Visualization:**
 - Explored fundamental characteristics of Iris species through descriptive statistics and interactive visualizations, revealing distinct clusters and correlations in petal and sepal measurements.
- **Species Differentiation:**
 - Identified clear patterns in petal length and width that differentiate Iris species, showcasing the dataset's utility in teaching classification techniques in machine learning.
- **Scalable Analytical Framework:**
 - Established a scalable analytical framework that can be adapted for further exploratory data analysis or expanded to include additional datasets and more complex machine learning models.
- **Future Enhancements:**
 - Suggested future work includes refining machine learning models to enhance species classification accuracy and integrating real-time data processing for dynamic analysis updates.

QUESTIONS?

